

AYMANE AIT DADS

Machine Learning Computer Vision Intern | PyTorch · CLIP ViT · TTA

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May 08, 2026

Hiring Team - Machine Learning Computer Vision Intern

Syntiant Corp. · Irvine, CA, USA

Dear Syntiant Hiring Team,

Syntiant's goal of expanding **computer vision detection capabilities through cutting-edge algorithm exploration** is exactly the problem I spent the last year solving: I ranked **#1 on the EURECOM private leaderboard** by fine-tuning **CLIP ViT-L/14** (428M parameters) on 250K images spanning 25+ generator types, designing a detection system robust enough to generalize far outside its training distribution.

Two results map directly to what your production pipeline needs. First, to ensure **out-of-distribution robustness** - a hard requirement when deploying detection at scale - I engineered a **10-view test-time augmentation** strategy that measurably improved inference reliability without retraining. Second, I ran structured **ablation experiments** across transformer block unfreezing depths (**4, 6, and 8 layers**) and learning rate schedules, proving that 1-epoch fine-tuning preserves generalization better than extended re-training - the kind of finding that directly informs production decisions.

Syntiant sits at a rare intersection: shipping AI at the semiconductor level where every inference cycle is constrained. My work with **FP16 mixed precision and GradScaler** on a T4 GPU reflects the same efficiency-first mindset. I would welcome the chance to discuss how I can contribute to your detection research this summer.

Sincerely,

AYMANE AIT DADS